Discussion Section Worksheet 5: Solutions

1. The height of an adult emperor penguin is normally distributed with a mean of 4.11 ft and a standard deviation of 0.18 ft.
   a) Define the random variable of interest, X.
      X = the height of an adult emperor penguin
   b) State the distribution of X.
      X is N(4.11, 0.18)
   c) What is the average height of adult emperor penguins?
      E(X) = 4.11 ft
   d) What is the probability that the height of an adult emperor penguin is at most 4.01 ft?
      $P(X \leq 4.01) = P(Z \leq -0.56) = 0.2877$
   e) What is the probability that the height of an adult emperor penguin exceeds 4.2 ft?
      $P(X > 4.2) = P(Z > 0.5) = 1 - 0.6915 = 0.3085$
   f) What is the probability that the height of an adult emperor penguin is between 3.73 and 4.24 ft?
      $P(3.73 \leq X \leq 4.24) = P(Z \leq 0.72) - P(Z < -2.11) = 0.7642 - 0.0174 = 0.7468$

2. A driver selects a sample of 11 gas stations in California and records their current price ($) for unleaded gasoline:

   2.599, 3.459, 2.899, 2.859, 3.699, 3.299, 3.199, 4.099, 3.999, 2.899, 2.999

   a) Find the mean, median, and mode(s).
      2.599, 2.859, 2.899, 2.899, 2.999, 3.199, 3.299, 3.459, 3.699, 3.999, 4.099
      Mean = 3.2735
      Median = 3.199
      Mode = 2.899
   b) Find the five number summary.
      Min = 2.599, Q1 = 2.899, Q2 = 3.199, Q3 = 3.699, Max = 4.099

3. In each situation, explain whether the value given in **BOLD** is a parameter or a statistic:
   a) The Center for Aviation calculates the mean age of all first-year flight students to be **28.7 years**.
      parameter
   b) A park ranger found the median weight of 25 bighorn sheep to be **271 pounds**.
      statistic

4. The length of a round stingray (urobatis halleri) is normally distributed with a mean of 19.2 inches and a variance of 5.29 inches^2.
   a) Define the random variable of interest.
      X = the length of a round stingray
   b) State the distribution of the RV of interest.
      $X \sim N(19.2, 2.3)$
   c) Calculate the probability that the length of a round stingray exceeds 18 inches.
      $P(X > 18) = P(Z > -0.52) = 0.6985$
   d) Calculate the probability that the length of a round stingray is between 16 and 19 inches.

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$P(16 \leq X \leq 19) = P(Z \leq -0.09) - P(Z < -1.39) = 0.4641 - 0.0823 = 0.3818$

e) The length of a round stingray is longer than 77% of all other round stingrays. What is its length?

$P(Z \leq 0.74) \approx 0.77$

$(x - 19.2)/2.3 = 0.74$

$x = 20.902$ inches

5. The length of an adult oceanic manta ray is normally distributed with a mean of 14.9 ft and a standard deviation of 3.65 ft. Forty-nine adult oceanic manta ray is selected at random, and we are interested in the mean length.

a) Define the random variable of interest.

$\overline{X}$ = the sample mean length of 49 adult oceanic manta rays

b) State the distribution of the RV of interest.

$\overline{X} \sim N(14.9, 0.5214)$

c) What is the probability that the mean length is between 12 and 16 ft?

$P(12 \leq \overline{X} \leq 16) = P(Z \leq 2.11) - P(Z < -5.56) = 0.9826 - 0 = 0.9826$

d) What is the probability that the mean length is at least 15.5 ft?

$P(\overline{X} \geq 15.5) = P(Z \geq 1.15) = 1 - 0.8749 = 0.1251$

e) What is the probability that the mean length is 14.9 ft?

$P(\overline{X} = 14.9) = 0$

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